

trodes are immersed in the aqueous solution of drug. By injecting currents from the lithium-ion battery, microbubbles of oxygen and hydrogen are generated due to the electrolysis reaction given below:

Anode: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2(\text{gas})$

Cathode: $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^- + \text{H}_2(\text{gas})$

Gases accumulate against the capping membrane until the membrane exceeds its rupture value (about 65 kPa)—a value when the membrane begins to rupture. Once the membrane ruptures, the drug is released. The lethal volume of air in an adult human is estimated to range from 200 to 300 ml. It is known that air bubbles of less than 30 μl dissolve harmlessly into the circulation. Since the volume of microbubbles generated by the device is about 120 nl—much smaller than 30 μl —the released bubble will not cause any health risk to the patient.

Sub-blocks of SoC

OOK receiver. Full integration and low power consumption are very important. Traditional direct-conversion and superheterodyne receivers consume a large amount of power and occupy a huge space. An alternative approach is adaptation of super-regenerative receiver and enveloped detection. Though this approach consumes less power, off-chip components, e.g., discrete inductor, bulk-acoustic-wave resonator and surface-acoustic-wave filter, occupy a huge space, posing a barrier to full system integration. Hence an envelope detection system based on the OOK receiver is adopted.

The envelope detection system based on the OOK receiver requires no off-chip components and on-chip inductor, enabling a reduction in size to 0.60 mm². The OOK receiver comprises a low-voltage amplifier, an envelope detector and a comparator with an output buffer as shown in Fig. 2. A simple self-biasing technique and envelope detection take place in this circuit. Neither a frequency synthesiser nor biasing circuits are required. The carrier frequency centered at 402 MHz is designed to comply with the medi-

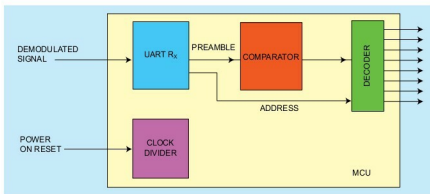


Fig. 4: Block diagram of MCU

cal implant communication system regulation for data transmission and reception in the body.

MCU. The function of the MCU is simplified to the basic requirements, i.e., register number recognition release command check and switch array control. Fig. 4 shows the architecture of the MCU. It consists of a clock divider, a decoder, a comparator and a universal asynchronous receiver/transmitter receiver (UART Rx). The UART Rx demodulates the received signal, converts serial bits to a parallel form, and sends the same to the comparator and the decoder for further reaction.

The received command sequence is composed of start/stop bits, address code and a preamble code. The address code is used to specify which branch at the decoder output will be injected with current. The preamble code is used as an activation key sent to a comparator so that the comparator can check the preamble against a default examining code. If these two codes are identical, an enable bit is sent to the decoder and current injection to the designated reservoir initiated according to the address code.

POR circuit. A POR circuit is implemented in the system to reset the register so that the previous values stored in the MCU registers do not activate drug release by mistake.

Regulator. The OOK receiver (1.8V) is biased by a voltage regulator, whereas the other sub-units of the SoC are directly biased by the battery. Controlled by the MCU, the regulator alters its output voltage and switches

the receiver between switching mode and quiet mode.

The regulator consists of a startup circuit, a bandgap reference and an error amplifier. The function of the bandgap reference is to generate a temperature-insensitive voltage. A start-up circuit is implemented to prevent the reluctance of the bandgap reference circuit to turn on by the supply voltage. The error amplifier, as a output stage of the regulator, uses a negative feedback to lock its output DC voltage to the reference voltage generated by the bandgap reference, making the output voltage very stable.

At drug-release activation, a DC voltage of 3V is applied to the electrodes, which then conduct a 2mA current into the solution. After activation, microbubbles occur and diffuse from the rupture of the membrane. Only 1.5 mA is needed to open the reservoir after the opening command is given. A Li-ion nanowire battery with an energy density of 7 mJ/mm³ and a button cell of 4.5mm diameter and 2mm height (32 cubic mm volume) can provide a total energy of about 403 mWh, enabling the device to power the system for 53 hours. The receiver is turned on for 1 μs every half second to conserve power.

Biocompatibility of the implant is very important. The surface of the device is made of silicon and silicon nitride, while for the packaging material titanium is used. These three materials are biocompatible. ●

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